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CLINICAL RESEARCH

Temporal Trends and Factors Associated With Cardiac Rehabilitation Participation Among Medicare Beneficiaries With Heart Failure

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ABSTRACT

OBJECTIVES The purpose of this study was to assess temporal trends and factors associated with cardiac rehabilitation (CR) enrollment and participation among Medicare beneficiaries after the 2014 Medicare coverage expansion.

BACKGROUND CR improves exercise capacity, quality of life, and clinical outcomes in heart failure (HF) with reduced ejection fraction (HFrEF). In 2014, Medicare coverage for CR was expanded to include chronic HFrEF.

METHODS Among Medicare beneficiaries from quarter (Q) 1 2014 to Q2 2016, 11,696 patients from 14,258 hospitalizations with primary discharge diagnosis of HF were identified. Patients with HF with preserved ejection fraction were excluded. Quarterly CR participation rates among hospitalized HF patients within 6 months of discharge were identified through outpatient administrative claims. The predictors of CR participation were assessed with the use of a multivariable logistic regression model that included patient- and hospital-level characteristics. A secondary analysis to assess participation rates of CR after outpatient encounters for HF was performed.

RESULTS Overall, only 611 (4.3%) and 349 (2.2%) eligible patients participated CR after primary hospitalization or outpatient visit for HF, respectively. There was a modest, statistically significant increase in CR participation after HF admissions (2.8% in Q1 2014; 5.0% in Q2 2016; $p < 0.001$) without significant increase after outpatient visits for HF (2.6% to 3.8%; $p = 0.21$). Younger age, male sex, nonblack race, previous cardiovascular procedures, and hospitalization at hospitals with available CR facilities were all independently associated with CR participation.

CONCLUSIONS CR participation among eligible Medicare beneficiaries with HFrEF was low with minimal increase since 2014 Medicare coverage decision. Sex, race, and institution-dependent variables were independent predictors of CR participation. (J Am Coll Cardiol HF 2021;9:471–81) © 2021 by the American College of Cardiology Foundation.

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the Author Center.

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ABBREVIATIONS AND ACRONYMS

CABG = coronary artery bypass graft

CMS = Centers for Medicare and Medicaid Services

CR = cardiac rehabilitation

CV = cardiovascular

HF = heart failure

HFpEF = heart failure with preserved ejection fraction

HFrEF = heart failure with reduced ejection fraction

ICD = International Classification of Diseases

MI = myocardial infarction

PCI = percutaneous coronary intervention

Heart failure (HF) hospitalizations are increasing in prevalence and are a major cause of morbidity, mortality, and health care costs (1). Exercise intolerance is the primary clinical manifestation of HF and is associated with poor quality of life and adverse clinical outcomes (2). Exercise training has been shown to improve exercise capacity and quality of life, and lower risk of hospitalization among patients with HF, particularly HF with reduced ejection fraction (HFrEF) (3-5). Current guidelines recommend cardiac rehabilitation (CR) for management of patients with stable HFrEF (Class 1B) (6). Furthermore, in 2014 the Centers for Medicare and Medicaid Services (CMS) approved supervised CR for patients with chronic stable HFrEF (7). However, there are few data available regarding enrollment and participation in CR in this key group of recently eligible high-risk patients. We examined the temporal trends and significant determinants of CR enrollment and participation among Medicare beneficiaries with a recent primary hospitalization or outpatient encounter for Medicare beneficiaries with chronic stable HFrEF.

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METHODS

STUDY DESIGN AND DATA SOURCE. We used a 5% national Medicare dataset from January 2014 through June 2016. This dataset is a 5% random unbiased sample of all Medicare beneficiaries over the period created by CMS. The data include information on the classification of Medicare eligibility, including both fee-for-service and managed care, and includes both Medicare Part A (inpatient) and Medicare Part B (outpatient) administrative claims for reimbursement. Carrier files include claims submitted by individual physicians. Outpatient and inpatient claims are captured separately. Primary, secondary, and comorbid conditions are recorded as International Classification of Diseases (ICD)-9th and -10th Edition codes. In addition, an accompanying annual record contains patient demographic details, including age, sex, and race or ethnicity. This also includes information on date of death throughout the study period. To examine the association between discharge hospital characteristics and participation in CR after discharge, we linked Medicare Hospital Identification Numbers to the American Hospital Association (AHA) Annual Survey, 2016, which included hospital demographic information and availability of ancillary hospital services.

SETTING AND PARTICIPANTS. HF hospitalization cohort. Our primary analysis included patients who had a recent admission with primary discharge diagnosis of HF (ICD-9: 428.x, 402.x1, 404.x1, 404.x3; ICD-10: 11.0, 13.0, 13.2) between the first quarter (Q1) of 2014 through Q2 of 2016 (Supplemental Table 1) and had continuous Medicare enrollment for 3 months previous and 6 months after the HF hospitalization. Exclusion criteria included death during the index hospitalization and discharge to skilled nursing facility, long-term acute care hospital, or hospice after the hospitalization. Because patients with HFpEF and patients within 6 weeks of a recent HF hospitalization are not eligible for CR coverage by the 2014 CMS coverage decision, patients with claims codes for HFpEF or re-hospitalization within 6 weeks of discharge were excluded. The final study cohort included 11,696 unique patients with 14,258 hospitalizations during the study period.

Outpatient HF cohort without hospitalization. The secondary cohort of interest was of outpatients with HF who were identified by 2 or more outpatient claims with a diagnosis code for HF in the primary position in 2 separate calendar quarters. CR enrollment and participation rates were assessed within 6 months of an outpatient HF visit. If a follow-up outpatient visit with primary HF diagnosis occurred within 6 months of the index visit, it was not included in the denominator for estimation of CR participation rates. Patients who had a hospitalization with primary diagnosis of HF during the study period were excluded from the outpatient cohort. Outpatients who did not have continuous Medicare enrollment 6 months after the second (confirmatory) outpatient visit with HF were also excluded. Mirroring the primary cohort, patients with HFpEF were excluded. The final outpatient population included 11,832 patients with 15,741 outpatient visits.

VARIABLES. We followed each hospitalization with primary diagnosis of HF and each outpatient visit with first diagnosis listed as HF for 6 months to assess enrollment and participation in CR. Consistent with previous approaches, the definition of CR participation was at least 1 claim with Current Procedure Terminology code 93797, 93798, G0422, G0423, or S9472 in outpatient-limited dataset files within either a 6-month follow-up period or before the next HF hospitalization (8). Patients were censored after enrollment and participation in CR or at death or loss to follow-up. Health care claims within 3 months before each hospitalization or outpatient visit were

analyzed to identify presence of cardiovascular (CV) and non-CV comorbidities, as detailed in Supplemental Table 2.

STATISTICAL ANALYSES. Demographic, clinical comorbidities, and index hospital characteristics were compared among patients who did versus did not participate in CR within 6 months after discharge or outpatient visit with the use of Pearson chi-square tests. The unit of analysis to assess temporal trends in CR was hospitalization with primary diagnosis of HF or outpatient visit with first diagnosis as HF. CR participation rates were calculated as proportion of HF hospitalizations or outpatient encounters (with primary diagnosis HF) within each quarter that were followed by a claim for CR over 6 months of follow-up. Linear regression was used to determine the significance of temporal changes in the participation rates. Subgroup analyses were performed to assess the temporal trends in the participation in CR across the following subgroups: men versus women, black versus nonblack patients, patients hospitalized in centers with versus without available CR, patients hospitalized in academic versus nonacademic centers, and patients hospitalized in centers with versus without cardiac surgery facilities.

Patient-level predictors of CR participation within 6 months of discharge were assessed in the overall study cohort by constructing a multivariable-adjusted logistic regression model with following covariates included in the model (model 1): the covariates of the model included age, sex, race, previous myocardial infarction (MI), atrial flutter/fibrillation, previous percutaneous coronary intervention (PCI) or coronary artery bypass graft (CABG), anemia, valvular surgery, cerebrovascular accident, hypertension, diabetes mellitus, diabetes mellitus with complications, peripheral vascular disease, renal insufficiency, chronic obstructive pulmonary disease, smoking, implantable cardioverter-defibrillator placement, and time (Q). Hospital-level predictors of CR participation were also assessed in a subset of the study participants ($n = 11,264$; 79%) with available linked data from the AHA survey on the relevant hospital-level characteristics. Within this subset, complete data on all key hospital-level characteristics was available for 10,017 patients (70.3%). The multivariable logistic model for identifying hospital-level predictors of CR participation included the following covariates (model 2): variables in model 1 + hospital-level characteristics (hospital teaching status, number of hospital beds, urban vs. rural location, and availability of CR) in the models. Generalized estimating equations (GEEs) were used to account for within-hospital clustering. In addition, we performed backward stepwise model selection on patient- and hospital-level characteristics based on quasilikelihood under the independence model criterion for each GEE model to determine the covariates for the model.

Sensitivity analyses were performed evaluating the temporal trends in CR participation among patients with a specific diagnosis code of HFrEF throughout the study period in both inpatient and outpatient study cohorts. HF subtype was determined through ICD-10 codes (HFpEF: 150.3; HFrEF: 150.2). An additional sensitivity analysis evaluating CR participation in a subset of patients with nonischemic etiology of HF as identified by exclusion of patients with administrative claims for PCI, CABG, or MI.

All statistics were performed with the use of R 3.6.2., with level of significance set at $p < 0.05$. The study protocol using deidentified CMS data for studying care patterns for HF was approved by the institutional review board at the UT Southwestern Medical Center. The requirement for informed consent was waived because the data were deidentified.

RESULTS

CR PARTICIPATION PATTERNS IN PATIENTS WITH HF. Of 11,696 HF patients with 14,258 hospitalizations for HF, only 611 (4.3%) participated in CR within 6 months of HF hospitalization, with lower participation in women (vs. men, 3.3% vs. 5%; $p < 0.001$) and black patients (vs. white patients, 2.9% vs. 4.7%; $p < 0.001$). At the hospital level, participation in CR following hospitalization was higher among patients hospitalized in teaching (vs. nonteaching) hospitals (6.1% vs. 3.4%; $p = 0.0025$), hospitals with (vs. without) available CR (5.0% vs. 2.0%, $p < 0.001$), and hospitals with (vs. without) cardiac surgery services (5.1% vs. 3.4%; $p < 0.001$). A greater proportion of patients discharged from urban hospitals had CR available at the discharging hospital and participated in CR (88% and 4.8%, respectively) versus those discharged from rural hospital (81% and 3.1%, respectively).

Among patients with outpatient encounters for primary HF diagnosis without a hospitalization event (11,832 patients with 15,741 visits), only 349 (2.2%) participated in CR within 6 months of the outpatient encounter. Participation was lower in women versus men (1.5% vs. 2.9%; $p < 0.001$), without differences across race-based subgroups (2.3% in white patients, 2.2% in black patients; $p = 0.51$).

Characteristics of patients with versus without CR participation in the HF hospitalization cohort are presented in **Table 1**. Patients who participated in at least 1 CR session were younger, more likely to have

TABLE 1 Baseline Characteristics of Patients With or Without Cardiac Rehabilitation Participation After Discharge From a Primary Heart Failure Hospitalization

	Cardiac Rehabilitation	No Cardiac Rehabilitation	p Value
Patient level (n = 11,696)	(n = 517)	(n = 11,179)	
Age, yrs	73 (67-79)	76 (68-84)	<0.001
Male	352 (68.1)	6,564 (58.7)	<0.001
Race			<0.001
White	443 (87.0)	8,678 (78.0)	
Black	52 (10.2)	1,866 (16.8)	
Other	14 (2.8)	586 (5.3)	
MI	210 (40.6)	3,640 (32.6)	<0.001
Atrial fibrillation	279 (54.0)	6,074 (54.3)	0.90
Hypertension	445 (86.1)	10,048 (89.9)	0.0067
Diabetes	258 (49.9)	5,668 (50.7)	0.76
Diabetes with complications	109 (21.1)	2,601 (23.3)	0.27
CKD	249 (48.2)	6,020 (53.9)	0.013
PVD	221 (42.7)	4,200 (37.6)	0.020
Anemia	102 (19.7)	2,867 (25.6)	0.0030
Lung disease	284 (54.9)	6,590 (58.9)	0.077
Transient ischemic attack	27 (5.2)	790 (7.1)	0.13
CVA	95 (18.4)	1,713 (15.3)	0.070
History of CABG	202 (39.1)	2,949 (26.4)	<0.001
History of PCI	114 (22.1)	1,968 (17.6)	0.012
History of valvular surgery	32 (6.2)	337 (3.0)	<0.001
Tobacco use	52 (10.1)	1,516 (13.6)	0.026
ICD placement	120 (23.2)	2,707 (24.2)	0.64
Hospital level (n = 9,225)*	(n = 425)	(n = 8,800)	
Region			<0.001
Northeast	72 (14.0)	1,916 (17.2)	
Midwest	184 (35.7)	2,689 (24.2)	
South	179 (34.7)	4,876 (43.8)	
West	81 (15.7)	1,647 (14.8)	
Teaching status			0.0025
Major	95 (22.0)	1,515 (17.2)	
Minor	232 (53.7)	4,537 (51.6)	
Non	105 (24.3)	2,741 (31.2)	
Rural hospital location	45 (10.4)	1,230 (14.0)	0.042
Total hospital beds	346 (188-537)	301 (178-511)	0.011
Cardiac catheterization availability	371 (94.9)	6,799 (86.6)	0.0015
Cardiac surgery availability	295 (75.4)	5,361 (68.3)	0.0035
Cardiac rehabilitation availability	371 (94.9)	6,799 (86.6)	<0.001

Values are median (interquartile range) or n (%).

CABG = coronary artery bypass graft; CKD = chronic kidney disease; CVA = cerebrovascular accident; ICD = implantable cardioverter-defibrillator; MI = myocardial infarction; IQR = interquartile range; PCI = percutaneous coronary intervention.

history of MI and CV procedures (CABG, PCI, valvular surgery), and less likely to have comorbidities such as hypertension, kidney disease, and lung disease. Findings were similar among patients with versus without CR participation in the outpatient cohort (Supplemental Table 3) and in the sensitivity analysis cohort of patients with specific diagnosis code for HFrEF in the post-discharge (Supplemental Table 4) and outpatient settings (Supplemental Table 5).

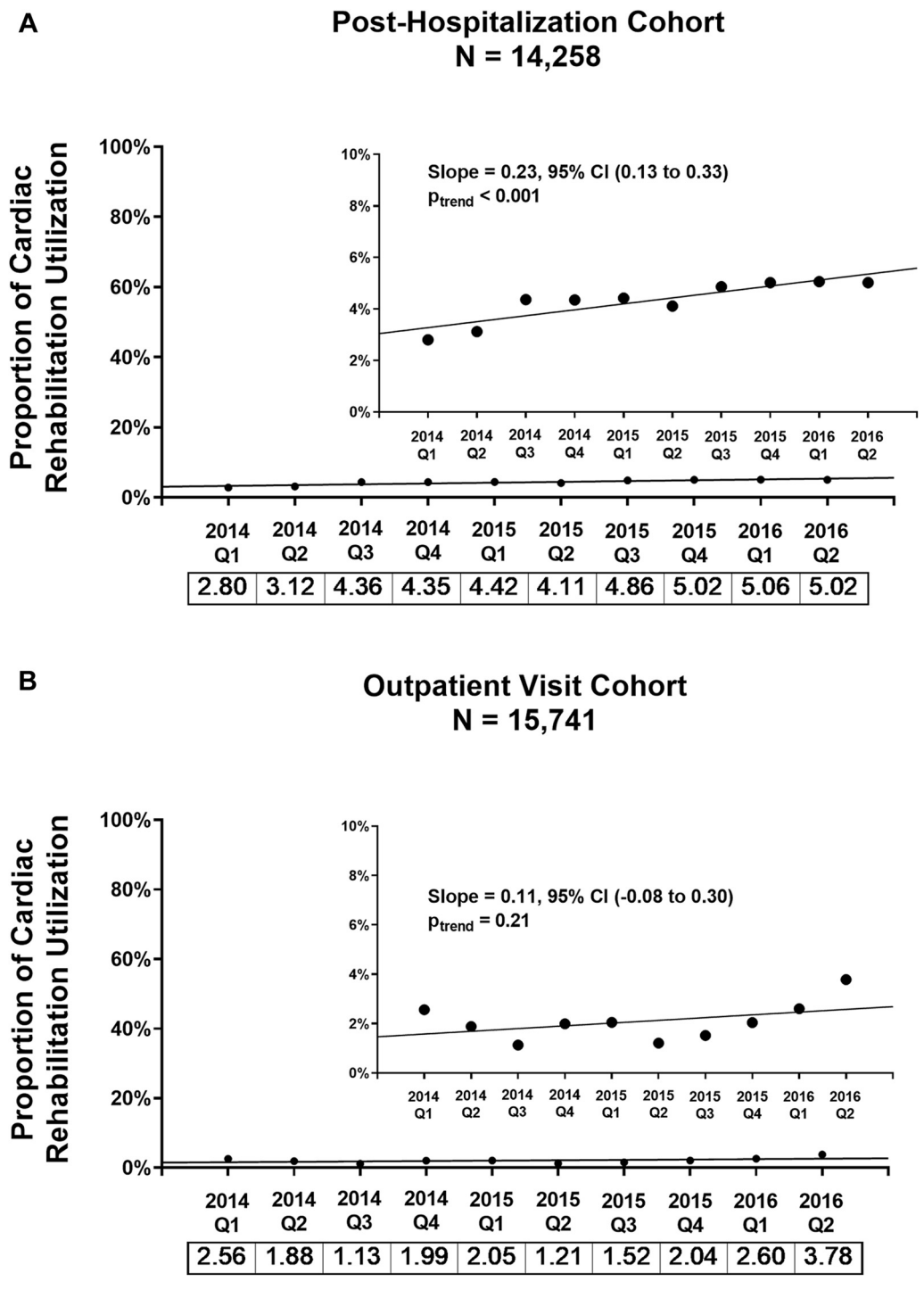
TEMPORAL TRENDS IN CR PARTICIPATION. Temporal trends in CR participation after HF hospitalization and outpatient visit for HF are shown in **Figure 1**. CR participation increased modestly over the study period (2.8% in Q1 2014 to 5.0% in Q2 2016; slope = 0.23; p for trend < 0.001) in the post-hospitalization cohort. A significant temporal increase in CR participation was noted among men (slope = 0.33; p for trend <0.001) but not in women (slope = 0.09; p for trend = 0.22) (**Figure 2**). There was no significant increase in CR participation over time for HF patients in the outpatient cohort (2.6% in Q1 2014 to 3.8% in Q2 2016; slope 0.11; p for rend = 0.21).

Across race-specific groups, Black and non-Black patients had temporal increases in participation of CR after hospitalization for HF (**Figure 3**). There were no race-based differences in CR participation for HF patients in the outpatient cohort. Among hospital-based subgroups, a significant temporal increase in CR participation was observed among patients discharged from academic hospitals, those with available CR, and those with cardiac surgery facilities (Supplemental Figure 1). No temporal change in CR participation after discharge was observed among patients discharged from hospitals without CR or cardiac surgery facilities and from non-academic-affiliated hospitals.

In sensitivity analysis, a low overall CR participation rate with significant temporal increase was also observed following hospitalizations with specific claim codes for HFrEF, similarly to the primary post-hospitalization study cohort (Supplemental Figure 2).

FACTORS ASSOCIATED WITH CR PARTICIPATION AFTER HF HOSPITALIZATION. In adjusted analysis, patient-level factors that were significantly associated with lower likelihood of CR participation were female sex, Black race, hypertension, anemia, and smoking (**Table 2**). Other comorbidities, such as lung disease, kidney disease, and presence of implantable cardioverter-defibrillator, also were associated with lower likelihood of CR participation with a trend toward statistical significance. In contrast, younger age, previous CABG, valvular surgery, and PCI were significantly associated with higher likelihood of CR participation (**Table 2**). Among hospital-level characteristics, hospitalizations at centers with academic affiliation and those with available CR were associated with a significantly higher likelihood of participation. Number of hospital beds was not significantly associated with likelihood of CR participation in adjusted analysis (**Table 2**). In a subset of patients with nonischemic etiology of HF, presence of comorbidities such as atrial fibrillation, hypertension, diabetes complications, and smoking were significantly associated with lower likelihood of CR participation. In contrast, younger age and presence of

FIGURE 1 Temporal Trends in Participation for Cardiac Rehabilitation Among Medicare Beneficiaries With Heart Failure



(A) Heart failure hospitalization cohort. **(B)** Heart failure outpatient cohort. The **numbers below the x-axis** show the proportional utilization of cardiac rehabilitation for each quarter.

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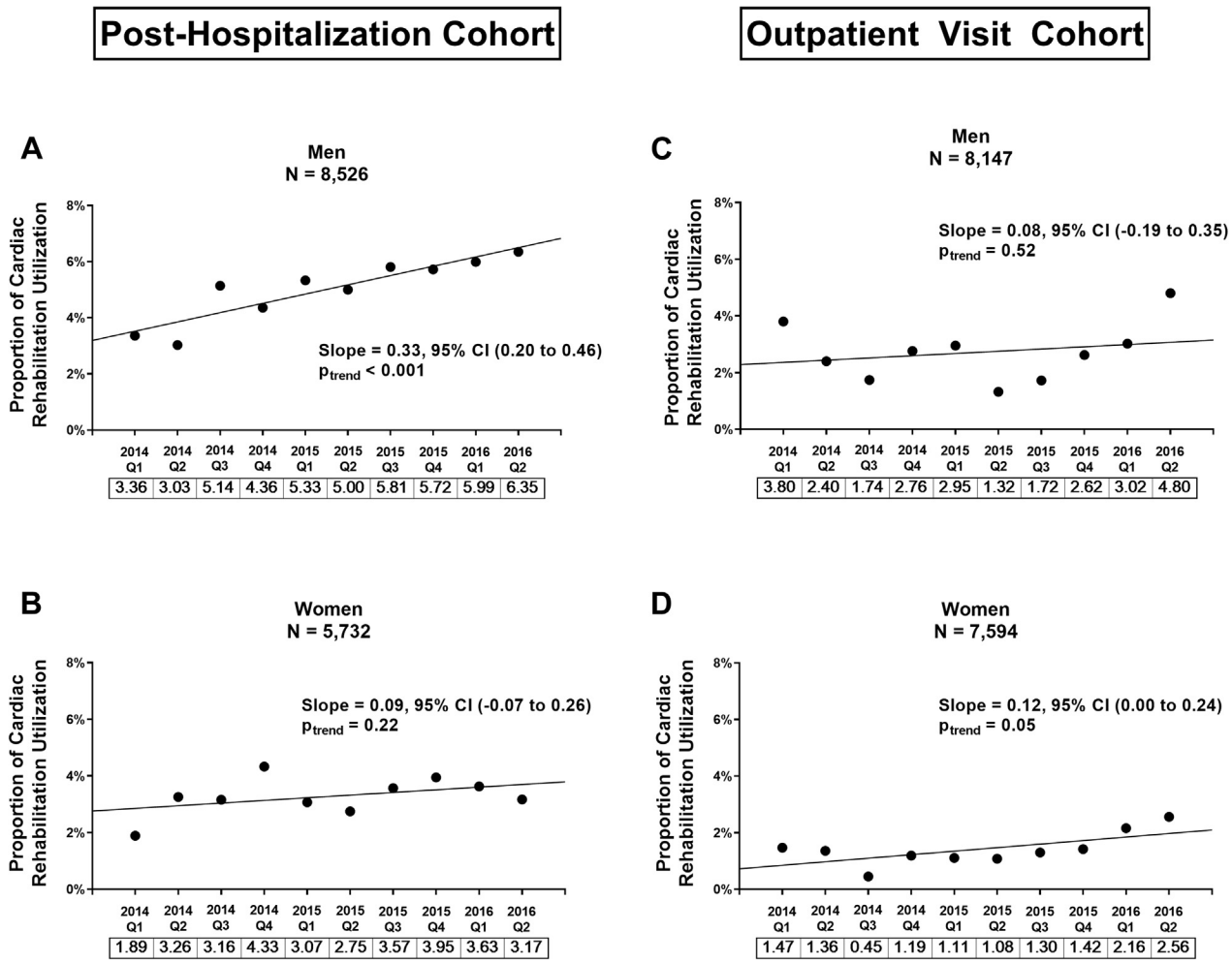
FIGURE 2 Sex-Stratified Temporal Trends in Participation for Cardiac Rehabilitation Among Medicare Beneficiaries With HF



(A) Men discharged after a primary heart failure (HF) hospitalization. **(B)** Women discharged after a primary HF hospitalization. **(C)** Men with HF seen in an outpatient visit without history of HF hospitalization. **(D)** Women with HF seen in an outpatient visit without history of HF hospitalization. The **numbers below the x-axis** show the proportional utilization of cardiac rehabilitation for each quarter.

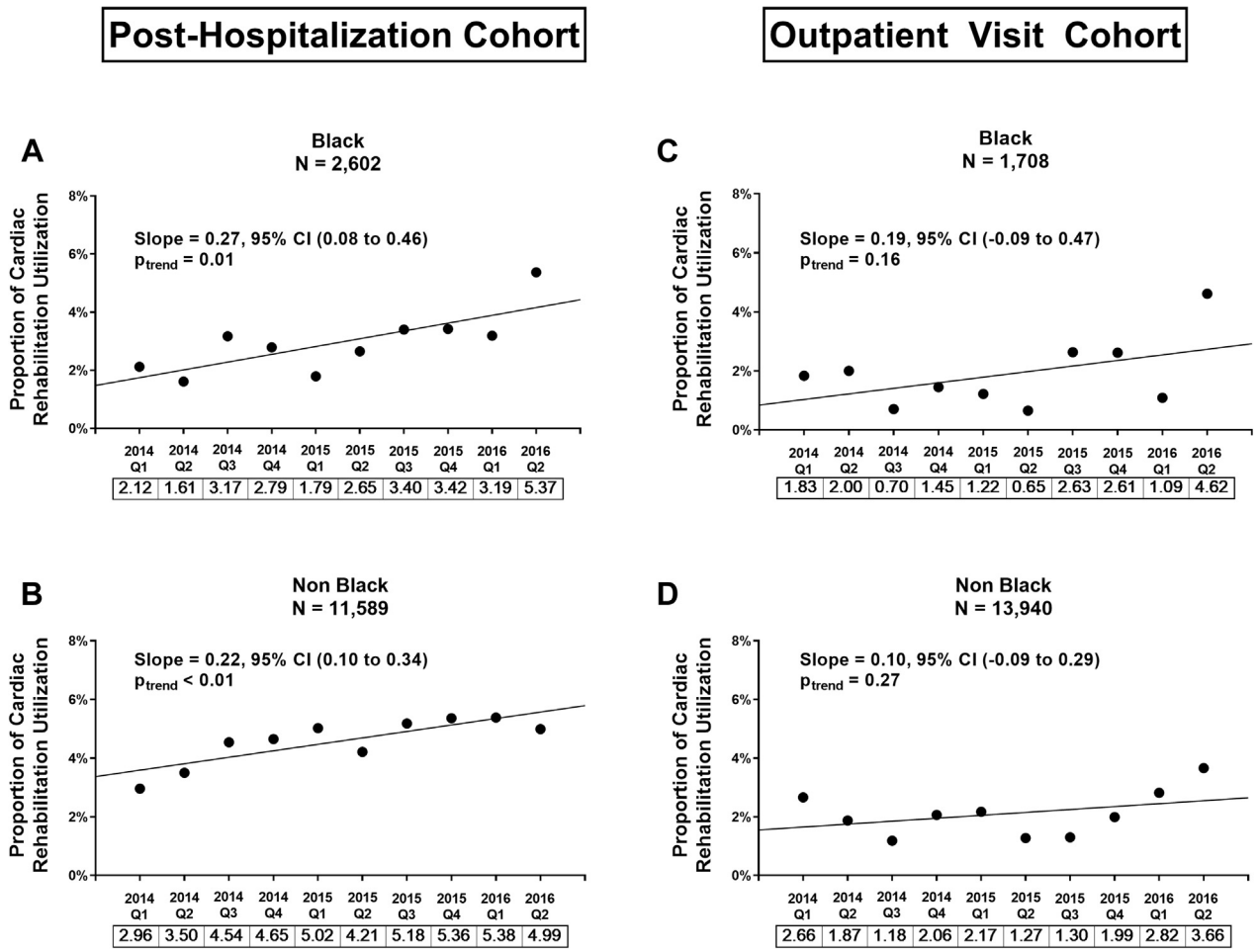
cerebrovascular diseases were significantly associated with greater likelihood of participation in CR (Supplemental Table 6).

DISCUSSION

In this analysis among Medicare beneficiaries with HF from 2014 to 2016 who were eligible for CR, we observed a very low rate of participation in CR after HF hospitalization or outpatient visit for HF. Furthermore, there were only very modest increases in CR participation in the >2 years following the 2014 Medicare coverage decision. Significant patient- and hospital-level predictors of CR participation included younger age, male sex, White race, history of CV procedures, and availability of CR at the index institution (**Central Illustration**). These data suggest marked systematic underparticipation in CR among patients with HF.

CR is an important component of the multidomain management of patients with CV diseases. In contrast with the low rates of CR participation among HF patients, previous studies have demonstrated substantially higher rates of CR participation among patients with other CV indications (Supplemental Table 7) (8–14). The low rates of CR participation in HF patients observed in the present study are consistent with some previous reports among Medicare beneficiaries and patients at Veterans Affairs hospitals (8,10). This study adds to the existing literature by

FIGURE 3 Race-Stratified Temporal Trends in Participation for Cardiac Rehabilitation Among Medicare Beneficiaries With HF



(A) Black patients discharged after a primary heart failure (HF) hospitalization. **(B)** Non-Black patients discharged after a primary HF hospitalization. **(C)** Black patients with HF seen in an outpatient visit without history of HF hospitalization. **(D)** Non-Black patients with HF seen in an outpatient visit without history of HF hospitalization. The **numbers below the x-axis** show the proportional utilization of cardiac rehabilitation for each quarter.

evaluating the CR participation after both HF hospitalizations and outpatient visits for HF, across different patient subgroups, and the temporal trends in CR since approval of coverage by CMS (7). We observed a modest but significant increase in participation in the overall cohort. While the trend in participation after hospitalization is encouraging, a substantial implementation gap remains in the uptake of CR among patients with HF.

Several factors may underlie the observed low rates of enrollment and participation in CR among patients with HF. First, patients with HF are usually older and have a higher burden of comorbidities and poor functional status, and therefore may be less likely to participate (15). Second, the referral rates for CR during HF hospitalization are low (16). Third, the lack of physician-perceived benefits from CR in HF may also contribute to lower participation. Fourth, CR coverage approved by CMS applies only to a relatively narrow group of patients and excludes those with HFpEF, who represent a majority of older HF patients and those with recent HF hospitalization, limiting the eligibility for participation in CR (17). There may also be a temporal lag in uptake after approval of CMS coverage, and future studies will be needed to determine if participation increases substantially to the rates observed for other indicated CV conditions.

At the patient level also, there are many barriers that may contribute to the low CR participation rates despite physician referral. Known barriers to CR include travel, willingness to participate, navigating the medical system, and poor communication. We

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TABLE 2 Patient and Hospital-Level Factors Associated With Cardiac Rehabilitation Participation Among Patients After Discharge From a Primary Heart Failure Hospitalization

	OR (95% CI)	p Value
Clinical (Model 1)		
Age	0.98 (0.97–0.98)	<0.001
Sex (female vs. male)	0.73 (0.60–0.89)	0.002
Race (Ref.: White)		
White	1.00	Ref.
Black	0.60 (0.44–0.82)	0.002
Other	0.40 (0.24–0.67)	<0.001
Atrial fibrillation	0.92 (0.76–1.52)	0.017
MI	1.26 (1.04–1.47)	0.11
Anemia	0.77 (0.62–0.96)	0.021
CVA	1.28 (1.02–1.60)	0.031
Hypertension	0.69 (0.54–0.89)	0.004
Diabetes	0.89 (0.73–1.09)	0.26
Diabetes with complications	0.82 (0.64–1.05)	0.11
Peripheral vascular disease	1.14 (0.94–1.39)	0.17
CKD	0.84 (0.69–1.03)	0.090
Lung disease	0.86 (0.71–1.03)	0.094
Tobacco use	0.55 (0.41–0.74)	<0.001
History of PCI	1.42 (1.14–1.78)	0.002
History of CABG	1.49 (1.23–1.80)	<0.001
History of valvular surgery	1.85 (1.25–2.74)	0.002
History of ICD	0.80 (0.64–1.00)	0.054
Hospital-level characteristics (Model 2)		
Urban vs. rural hospital	1.37 (0.96–1.94)	0.083
Number of hospital beds	1.00 (1.00–1.00)	0.53
Availability of cardiac rehabilitation	2.58 (1.60–4.15)	<0.001
Teaching affiliation (Ref.: major)		
Minor	0.76 (0.56–1.03)	0.077
Non	0.64 (0.44–0.92)	0.017

Model 1: multivariable logistic regression analysis with only patient-level demographic and clinical variables (25 patients were excluded owing to missingness of race; n = 14,191). Model 2: multivariable logistic regression analysis with both patient-level variables in model 1 and hospital-level characteristics presented in this table (25 patients were removed owing to missingness of race, and 4,174 patients were excluded owing to missingness of hospital-level information; n = 10,017).

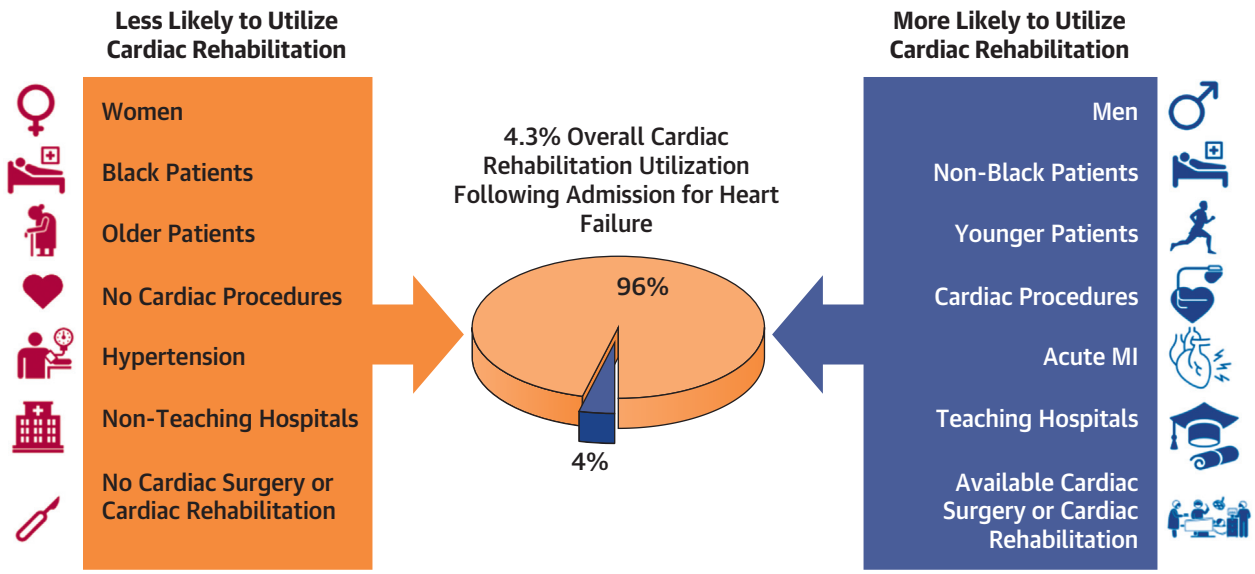
CI = confidence interval; OR = odds ratio; other abbreviations as in Table 1.

observed significant differences in baseline characteristics of patients with (vs. without) CR participation after discharge from a primary HF hospitalization. Consistent with previous observations for patients with other CV indications for CR, older patients, women, and Black patients were all less likely to participate in CR after a HF hospitalization (9,14). Previous studies have demonstrated that logistical challenges such as access to transportation and work or family responsibilities may pose a greater barrier to participation in CR for women compared with men (18). Similarly, older patients, who have a higher burden of exercise intolerance and HF symptoms may face greater barriers to CR participation. These include lack of confidence in ability to self-manage their condition, perception of exercise as tiring or painful, higher burden of comorbidities and frailty, and lack of physician encouragement (6,19). We also observed lower CR participation among racial minorities, highlighting race disparities in HF management (20). Consistent with the racial disparities in access and utilization in other health care domains (21–24), low CR participation rates among Black patients may be related to both patient-specific factors and societal/structural racism. Black patients have higher readmission rates for HF, lower post-discharge follow-up, and get referred for CR less often (16,25,26). Furthermore, because patients have copayments billed per session of CR, socioeconomically disadvantaged Black patients may face significant financial barriers to support enrollment and participation. Finally, Black communities may have poor access to rehabilitation owing to a lack of CR centers in their vicinity and inadequate transportation options. Further investigation is needed into the cause of these race-based disparities and the possible role of both implicit and explicit biases in CR enrollment and participation.

Disparities were also seen at the hospital level, with significantly lower rates of CR participation in patients discharged from nonteaching hospitals and hospitals without available cardiac surgery, cardiac catheterization, and CR facilities. Patients discharged from rural (vs. urban) hospitals also had lower rates of CR participation, which may be related to less availability of CR in rural hospitals. These system-level factors may also speak to geographic difficulties faced by patients with greater distances to travel for specialized cardiac care and a relative scarcity of CR centers (27). These findings also highlight the need for advocacy in the support of the hospitals who do not have enough resources, such as those providing care in rural regions, even as they care for patients at the highest risk for poor outcomes.

The present study findings have important clinical and health policy implications. CR is a key management strategy in the care of patients with HF and has been shown to improve exercise capacity, quality of life, and clinical outcomes (3,4,6,28). The findings of overall low participation rates and differences in participation across sex- and race-based subgroups of patients with HF highlights the existing implementation gap and disparities in utilization of this guideline-recommended therapy. Several potential strategies may improve participation in CR among patients with HF. First, education of physicians and patients about the guideline recommendations and benefits of CR may increase overall referral rates. Second, physician recognition of disparities in CR

CENTRAL ILLUSTRATION Rates and Predictors of Participation in Cardiac Rehabilitation Among Patients With Heart Failure



Pandey, A. et al. J Am Coll Cardiol HF. 2021;9(7):471–81.

Female sex, Black race, older age, admission to a nonteaching hospital, and lack of cardiac surgery or cardiac rehabilitation facilities at the admitting hospital were some key factors associated with lower participation in cardiac rehabilitation post discharge. MI = myocardial infarction.

participation between different races and sexes may allow for improved dialog and counseling regarding sex- or race-specific barriers to CR participation. Third, incorporation of innovative approaches to CR including home-based CR or tele-rehabilitation, or m-health based rehabilitation, may help to address some of the logistical challenges to site-based supervised CR (29). Finally, broadening the existing criteria for CR referral for HF indication with emergence of new evidence regarding safety and efficacy of exercise training may further promote uptake of CR in HF patients. These changes may include initiation of CR during hospitalization or soon after discharge and expanding the CR eligibility in patients with HF to include those with mildly reduced EF or preserved EF (28,30).

STUDY LIMITATIONS. First, the data source was derived from Medicare administrative claims, which limits the population to mostly older adults. In addition, claims data lack granular clinical information. As such, some patients included in the calculation of participation rates may not have been eligible for CR owing to lack of physician-perceived benefit or severe comorbidities that may preclude CR participation. This lack of granular clinical data also limited the discernment of HF subtypes because this information was only available for part of the cohort. However, in sensitivity analysis, our findings in the subgroup with specific diagnosis codes for HFrEF were similar to findings in the overall cohorts after exclusion of those with diagnosis codes for HFpEF. Second, the use of a 5% Medicare sample may have led to unequal random missingness of individual patient claims data that may have affected the calculation of participation rates at smaller hospitals, which may have affected hospital-level results. However, the strength of this dataset is in its ability to assess CR participation and not only receipt of referral. Third, owing to availability of the data, the follow-up period of our study was through 2016. As CMS data for more recent years becomes available, future studies with a longer study period may provide more insights into changes in CR participation patterns. Fourth, the hospital-level characteristics, which were obtained by linking the 5% CMS cohort data with the AHA survey data, were missing in 20% to 30% of the study cohort because not all hospitals participate in the survey and the data reporting is not complete for each specific hospital characteristic. Therefore, our observations pertaining to the hospital-level predictors of CR participation may not be generalizable to all patients. Fifth, we excluded

patients discharged to skilled nursing facilities and long-term acute care hospitals because they may not be considered medically or physically capable of CR. Owing to these exclusions, we may have missed out on a small proportion of such patients who could have recovered and were referred for CR as an outpatient after discharge from a skilled nursing facility or long-term acute care hospital. Finally, because the number of patients who participated in CR was very low, we had limited statistical power to evaluate the patterns and predictors of substantial utilization of CR (>12 sessions) and differences in clinical outcomes among patients with different levels of CR participation (31). This is particularly relevant because the impact of CR on outcomes is proportional to the exercise volume, which is correlated with the number of sessions attended (5). Future studies using larger cohorts of CMS-linked HF patients or longer follow-up may help to address these knowledge gaps.

CONCLUSIONS

CR is markedly underutilized among eligible Medicare beneficiaries with chronic stable HFrEF and showed minimal increase in enrollment and participation during the first 2 years following the decision that extended Medicare coverage to this high-risk population. Participation was systematically lower among women and Black patients, highlighting existing disparities in the use of CR.

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PERSPECTIVES

COMPETENCY IN MEDICAL KNOWLEDGE: Cardiac rehabilitation in chronic heart failure can improve patient quality of life and exercise tolerance, but participation is low. Health professionals should refer patients with stable heart failure with reduced ejection fraction to cardiac rehabilitation and educate patients on potential benefits.

TRANSLATIONAL OUTLOOK: Both system-level and patient-level strategies and innovations are needed to improve utilization of cardiac rehabilitation in patients with heart failure with reduced ejection fraction.

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APPENDIX For supplemental figures and tables, please see the online version of this paper.

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